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224515(24)/024714(24)

Dip. in Engg. (Fifth Semester)/
P. T. D. C. (Seventh Semester)
EXAMINATION, Dec.-Jan., 2013-14
(Elect.)

ELECTRICAL MACHINES – II

Time : Three Hours] [*Maximum Marks : 100*
[*Minimum Pass Marks : 35*

Note : All questions are compulsory, unless mentioned otherwise.

1. Write the difference between A. C. and D. C. machines. 5

2. Describe the construction of A. C. machines. 10

Or

Describe stator and rotor windings of A. C. machines.

3. (a) Explain with diagram working of brushless alternator. 10

Or

Derive e. m. f. equation of alternator.

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(b) With a neat diagram explain construction of an alternator. 10

Or

Explain different excitation systems used in alternator.

(c) A 12-pole, 3-phase, star-connected alternator has 72 slots. The flux per pole is 0·0988 Wb. Calculate : 10

(i) the speed of rotation if the frequency of the generated e. m. f. is 50 Hz.

(ii) the terminal e. m. f. for full pitch coils and 8 conductors per slot.

(iii) the terminal e. m. f. if the coil span is reduced to 2/3 of the pole pitch.

Or

A 50 Hz, 600 r. p. m., salient pole alternator has a sinusoidal flux density having a maximum value of 1 Wb/m^2. The alternator has 180 slots wound with 2 layer 3 turns coil. The coil span is 15 slots and phase spread is 60°. The armature diameter is 1·25 m and core length 0·45 m. Find :

(i) Flux/pole.

(ii) Pitch factor and distribution factor.

(iii) Peak valve of e. m. f. per conductor.

(iv) Peak valve of e. m. f. per coil.

(v) r. m. s. phase and line voltage if machine is star-connected.

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4. (a) Derive torque equation and also condition for maximum torque during running condition of an induction motor. 10

(b) Explain with neat diagram construction and principle of operation of induction motor. 10

Or

Explain the concepts of rotating magnetic field.

(c) Explain speed control of induction motor with diagram. 10

Or

Explain direct on line and star-delta starter method used in induction motor.

(d) Draw and explain torque slip characteristic of an induction motor. 5

Or

A 500 h. p., 3-phase, 440 V, 50 Hz induction motor has a speed of 950 r. p. m. on full load. The machines has 6 poles. Calculate the full-load slip. How many cycles will the rotor voltage make per minute ?

5. Explain any *four* of the following : 5 each

(a) Shaded pole induction motor

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(b) A. C. series or universal motor

(c) Capacitor start and capacitor run motor

(d) Single-phase repulsion motor

(e) Reluctance motor

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